deeplearning.ai

Hyperparameter tuning

Tuning process

Hyperparameters

$\rightarrow \boxed{\alpha}$

$\boxed{\beta}$ ~0.9

$\beta_1, \beta_2, \varepsilon$

$0.9 \quad 0.999 \quad 10^{-8}$

$\boxed{\text{\# layers}}$

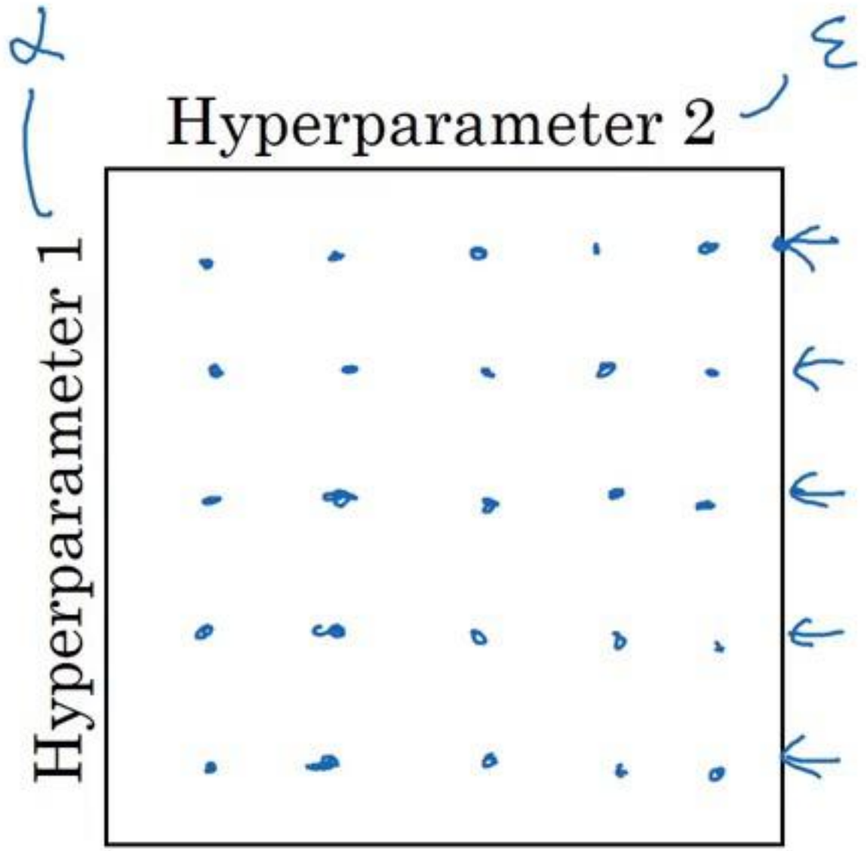
$\boxed{\text{\# hidden units}}$

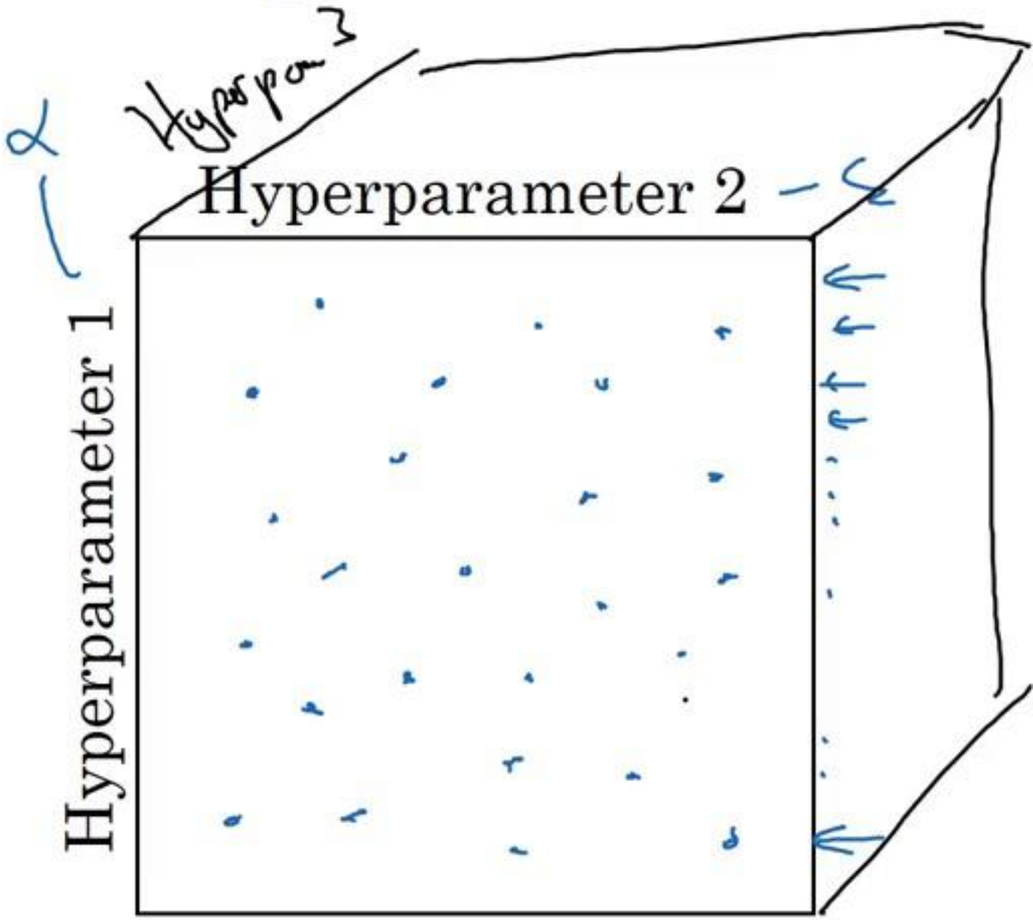
$\boxed{\text{learning rate decay}}$

$\boxed{\text{mini-batch size}}$

Andrew Ng

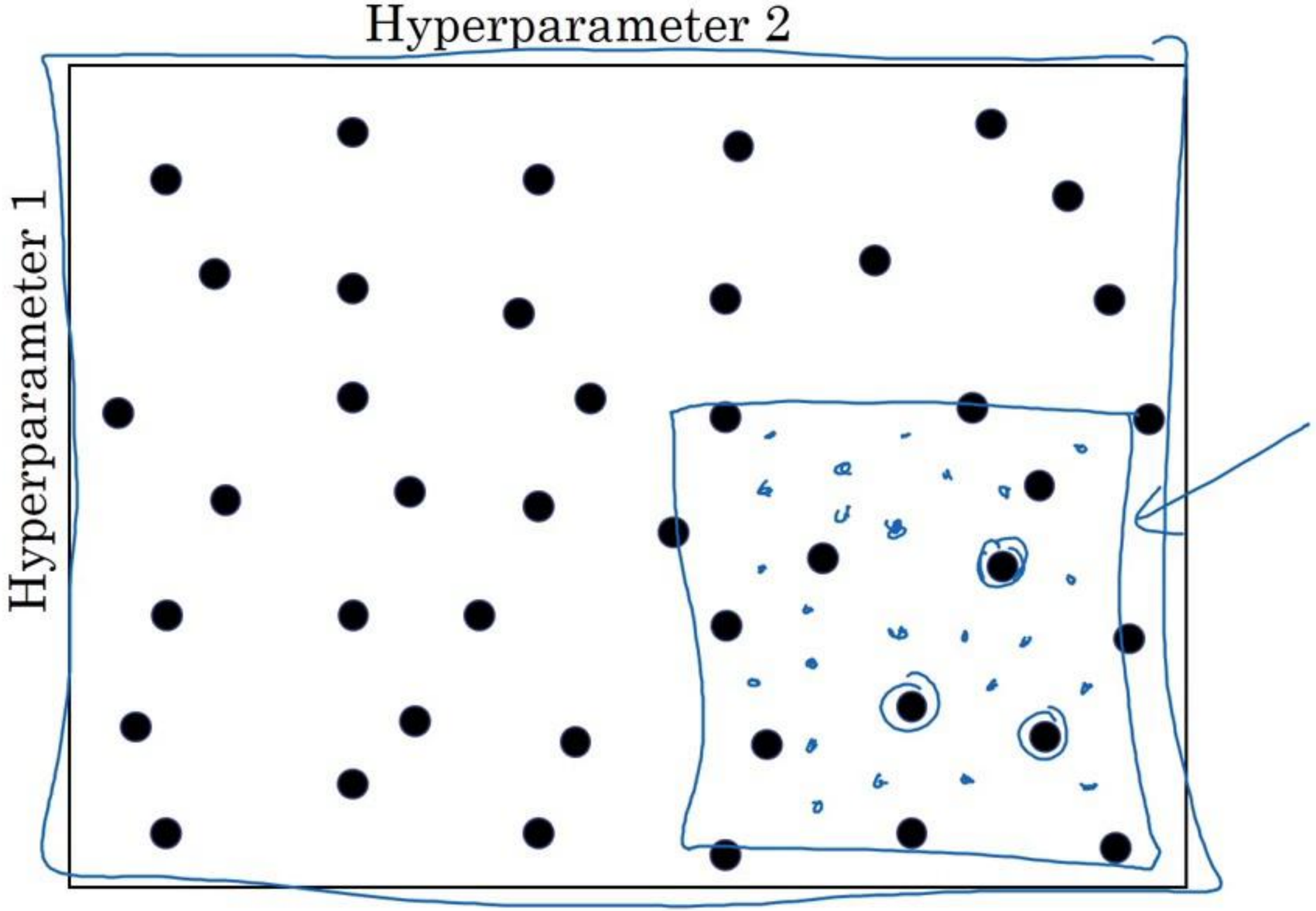
Try random values: Don't use a grid





Andrew Ng

Coarse to fine



Andrew Ng

deeplearning.ai

Hyperparameter tuning

Using an appropriate scale to pick hyperparameters

Picking hyperparameters at random

→ $n^{[l]} = 50, \ldots, 100$

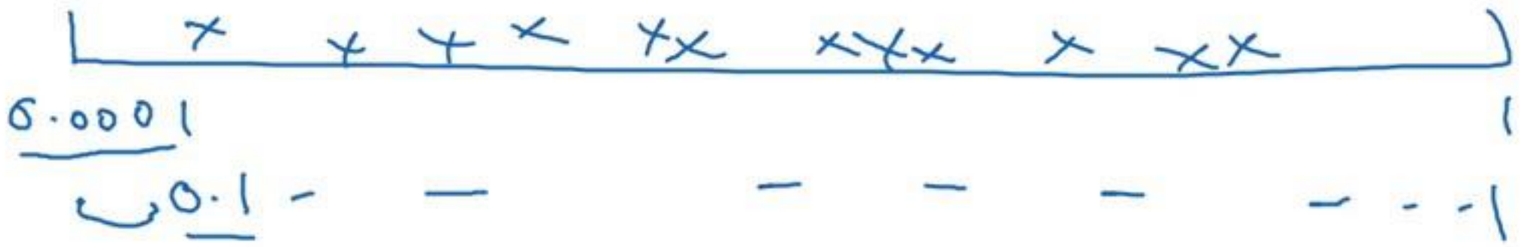
→ #layers L: 2 – 4

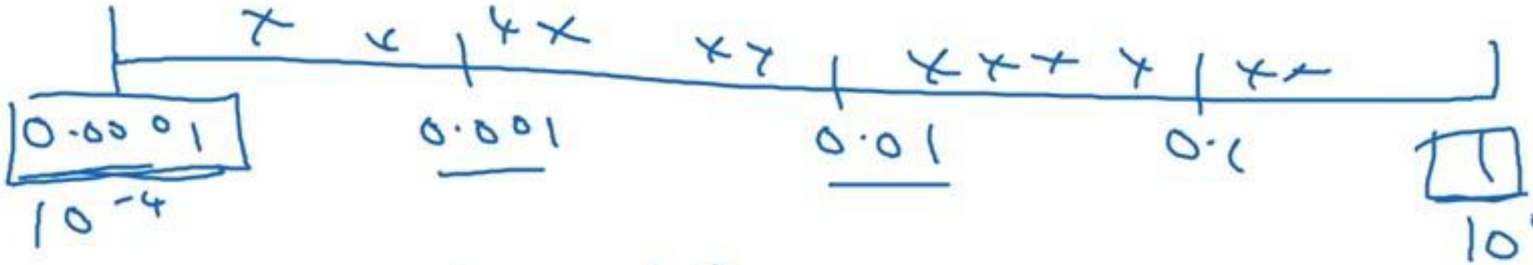
2, 3, 4

Andrew Ng

Appropriate scale for hyperparameters

$\alpha = 0.0001, \ldots, 1$





10^a

$a = \log_{10} 0.0001 = -4$

$r = -4 * np.random.rand()$ $\leftarrow$ $r \in [-4, 0]$

$\alpha = 10^r$ $\leftarrow$ $10^{-4} \ldots 10^{0}$

10^b

$b = \log_{10} 1 = 0$

$10^a \ldots 10^b$ $\quad$ $r \in [a, b]$ $\quad$ $\alpha = 10^r$

$[-4, 0]$

Andrew Ng

Hyperparameters for exponentially weighted averages

$\beta = 0.9 \quad \ldots \quad 0.999$

$\downarrow \qquad\qquad \downarrow$

$10 \qquad\qquad 1000$

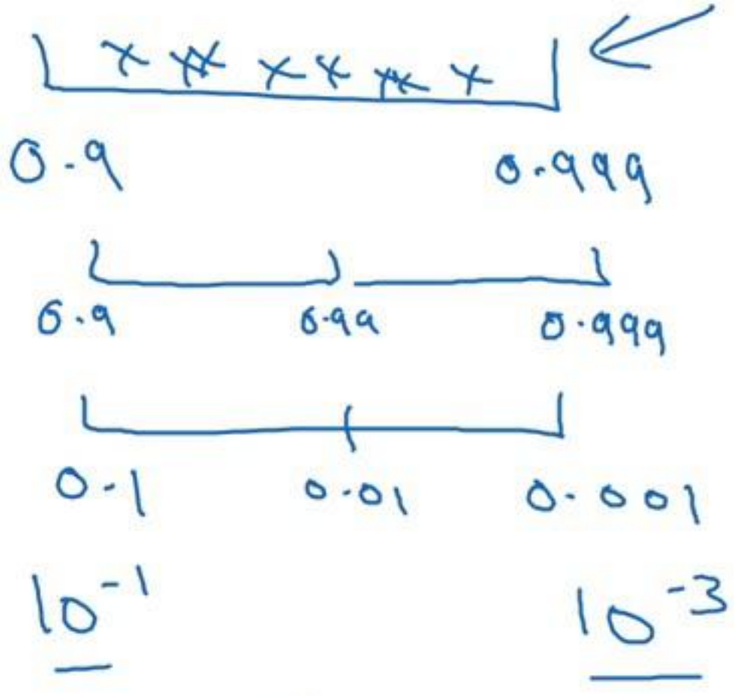
$1-\beta = 0.1 \quad \ldots \quad 0.001$

$\beta: \; 0.9000 \rightarrow 0.9005 \; \} \sim 10$

$\beta: \; 0.999 \rightarrow 0.9995$

$\sim 1000 \qquad \sim 2000$

$\frac{1}{1-\beta}$



$r \in [-3, -1]$

$1-\beta = 10^{r}$

$\beta = 1 - 10^{r}$

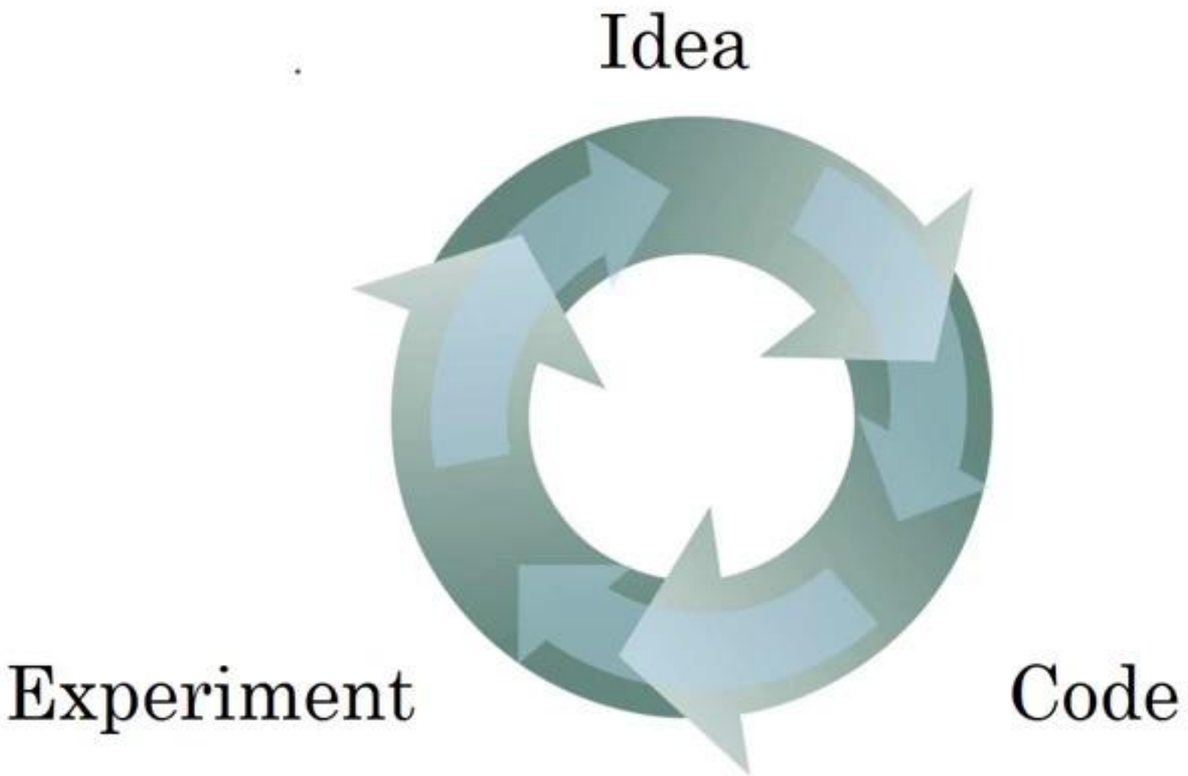
Andrew Ng

deeplearning.ai

Hyperparameters tuning

Hyperparameters tuning in practice: Pandas vs. Caviar

Re-test hyperparameters occasionally

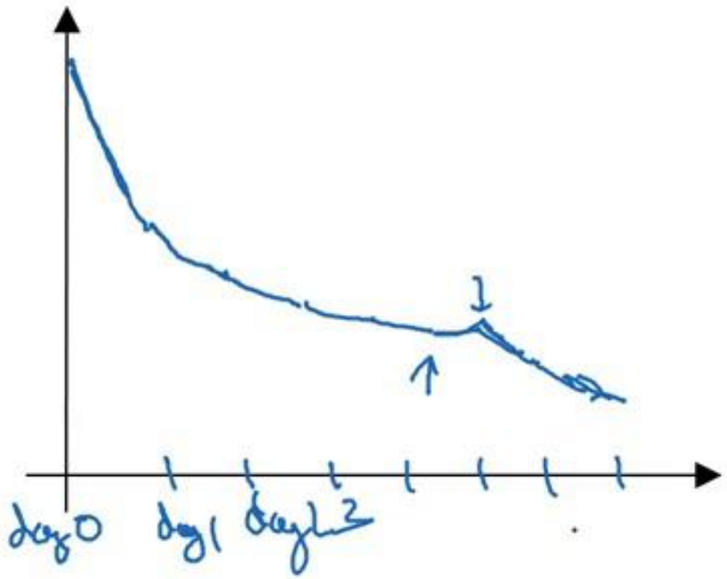


- NLP, Vision, Speech, Ads, logistics, ...

- Intuitions do get stale. Re-evaluate occasionally.

Andrew Ng

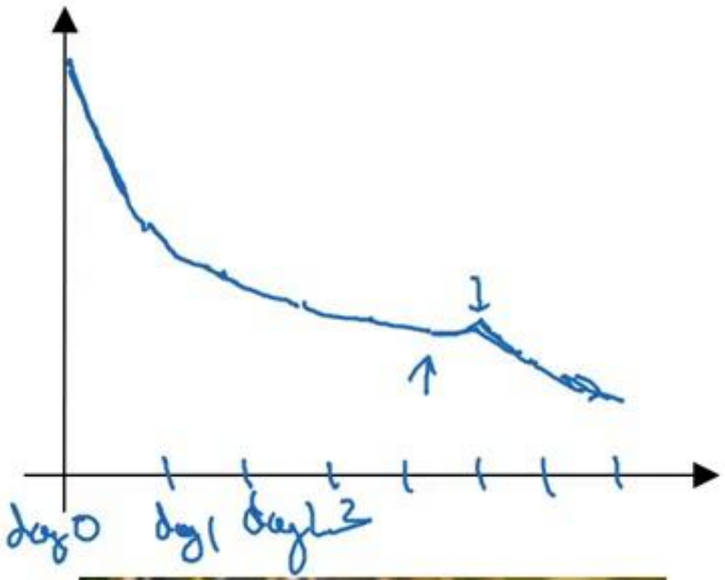
Babysitting one model



Training many models in parallel

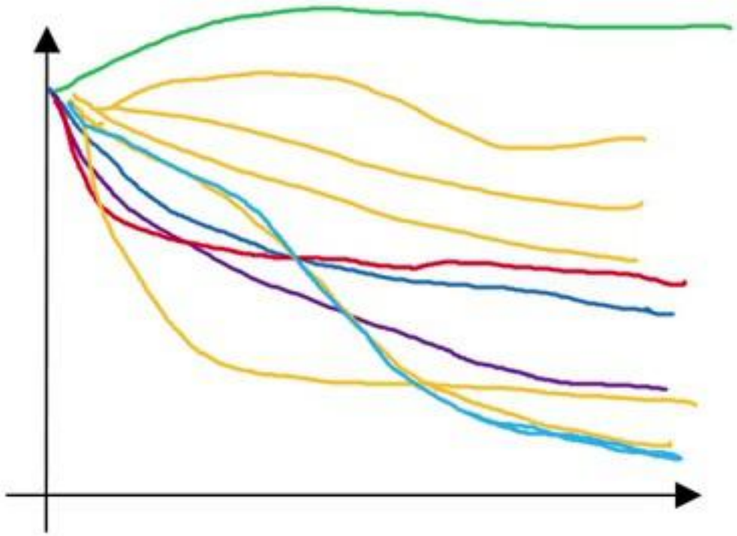
Andrew Ng

Babysitting one model



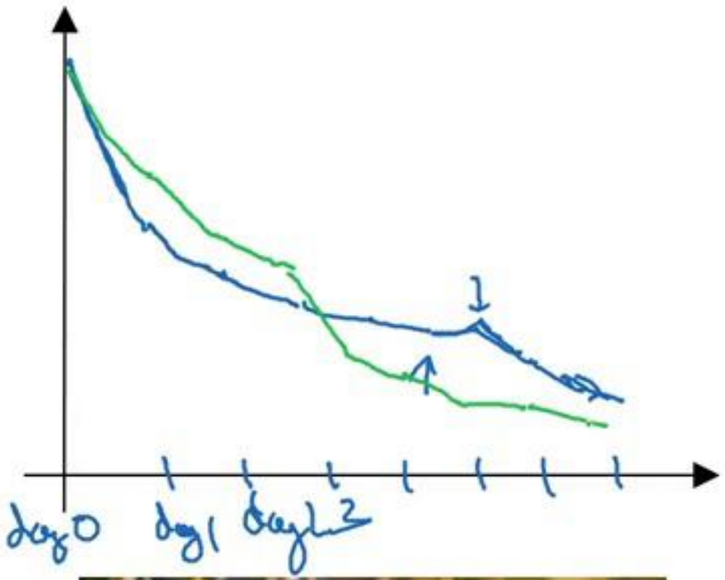
Panda

Training many models in parallel



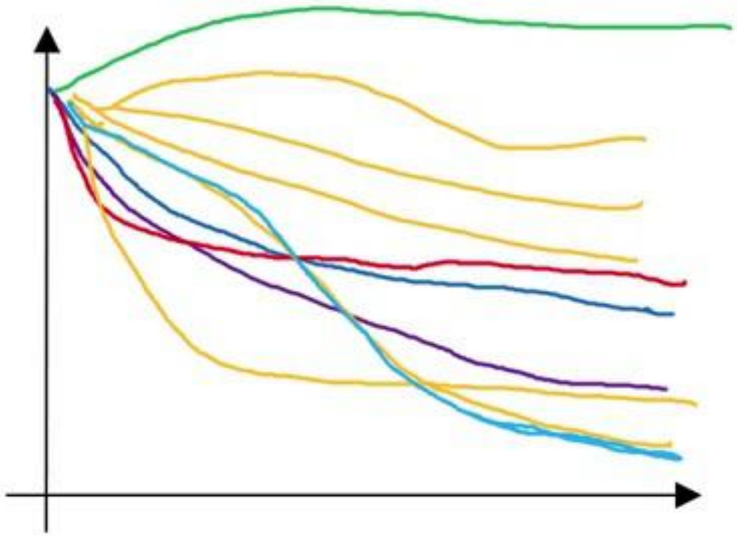
Andrew Ng

Babysitting one model



Panda

Training many models in parallel



Caviar

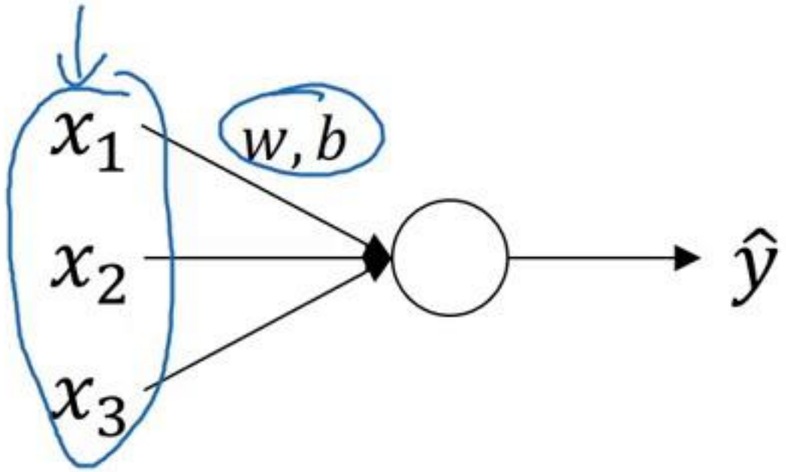
Andrew Ng

deeplearning.ai

Batch Normalization

Normalizing activations in a network

Normalizing inputs to speed up learning

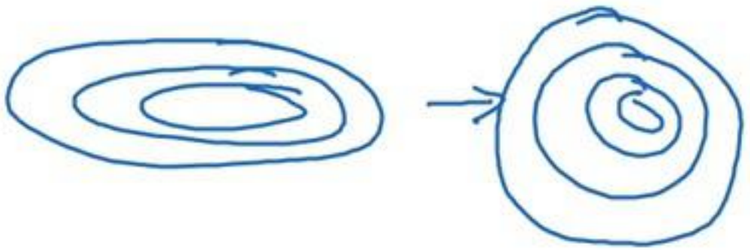


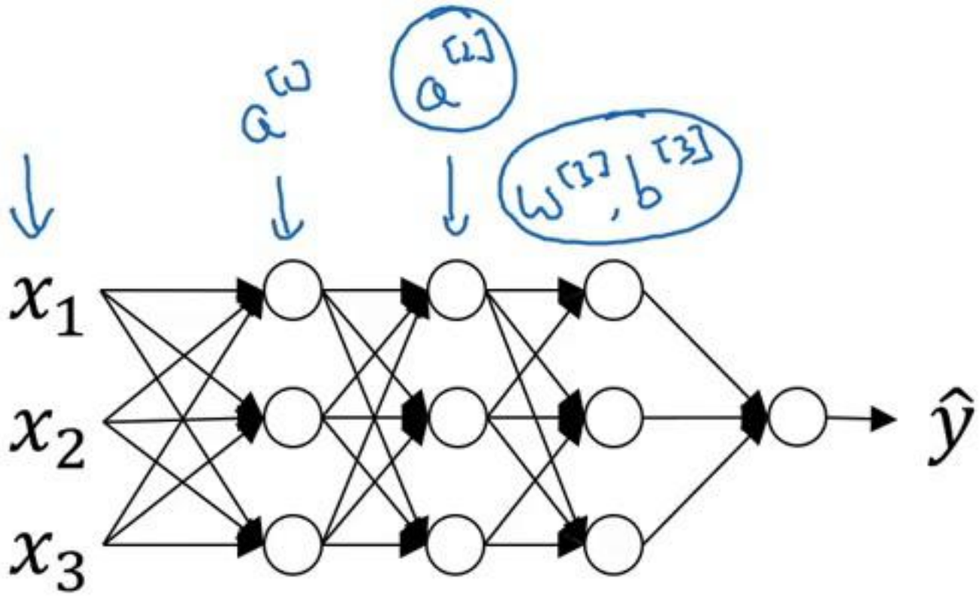
$$\mu = \frac{1}{m} \sum_i x^{(i)}$$

$$X = X - \mu$$ ← element-wise

$$\sigma^2 = \frac{1}{m} \sum_i x^{(i)2}$$

$$X = X / \sigma^2$$





Can we normalize $a^{[2]}$ so as to train $W^{[3]}, b^{[3]}$ faster

Normalize $z^{[2]}$

Andrew Ng

Implementing Batch Norm

Given some intermediate values in NN $\underbrace{z^{(1)}, \ldots, z^{(m)}}_{z^{[l](i)}}$

$$\mu = \frac{1}{m} \sum_i z^{(i)}$$

$$\sigma^2 = \frac{1}{m} \sum_i (z_i - \mu)^2$$

$$z^{(i)}_{norm} = \frac{z^{(i)} - \mu}{\sqrt{\sigma^2 + \varepsilon}}$$

$$\tilde{z}^{(i)} = \gamma z^{(i)}_{norm} + \beta$$

learnable parameters of model.

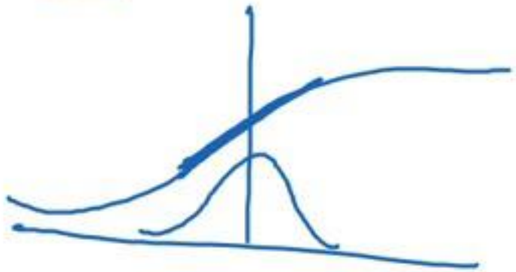
If $\gamma = \sqrt{\sigma^2 + \varepsilon}$

$\beta = \mu$

then $\tilde{z}^{(i)} = z^{(i)}$

X

$z^{(i)}$



Use $\tilde{z}^{[l](i)}$ instead of $z^{[l](i)}$.

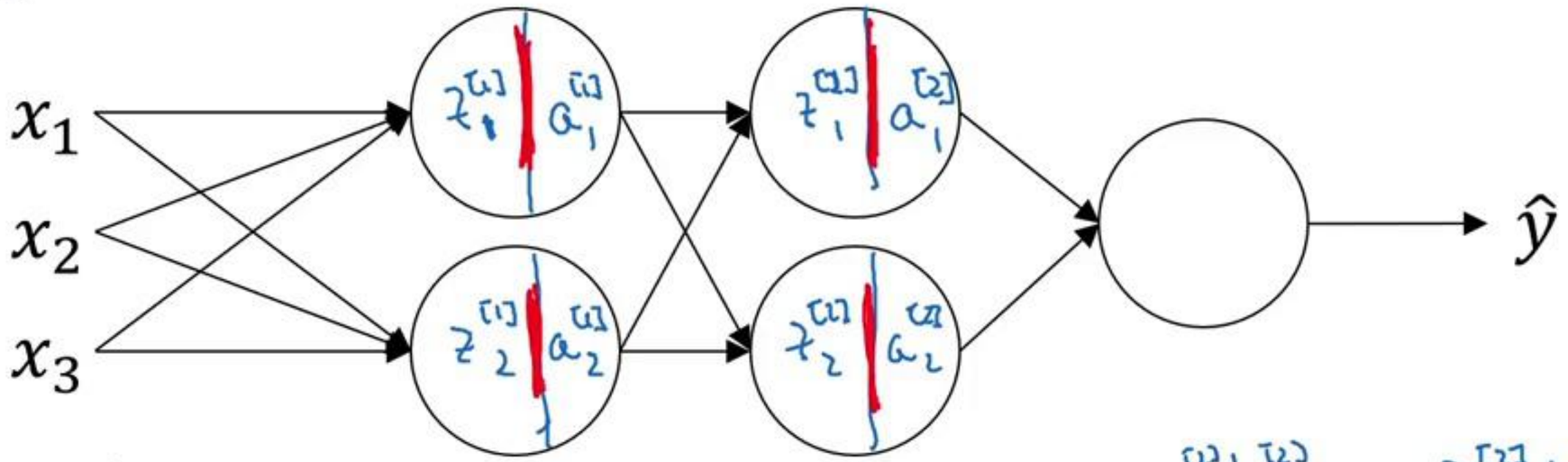
Andrew Ng

deeplearning.ai

Batch Normalization

Fitting Batch Norm into a neural network

Adding Batch Norm to a network



$$X \xrightarrow{W^{[1]}, b^{[1]}} z^{[1]} \xrightarrow[\text{Batch Norm (BN)}]{\beta^{[1]}, \gamma^{[1]}} \tilde{z}^{[1]} \rightarrow a^{[1]} = g^{[1]}(\tilde{z}^{[1]}) \xrightarrow{W^{[2]}, b^{[2]}} z^{[2]} \xrightarrow[\text{BN}]{\beta^{[2]}, \gamma^{[2]}} \tilde{z}^{[2]} \rightarrow a^{[2]} \rightarrow \cdots$$

Parameters: $W^{[1]}, \cancel{b^{[1]}}, W^{[2]}, \cancel{b^{[2]}}, \ldots, W^{[L]}, \cancel{b^{[L]}},$

$\rightarrow \beta^{[1]}, \gamma^{[1]}, \beta^{[2]}, \gamma^{[2]}, \ldots, \beta^{[L]}, \gamma^{[L]}$

$\rightarrow \beta$

$d\beta^{[l]}$ $\qquad \beta^{[l]} = \beta^{[l]} - \alpha\, d\beta^{[l]}$

tf.nn.batch_normalization $\leftarrow$

Andrew Ng

Working with mini-batches

$$X^{\{1\}} \xrightarrow{W^{[1]}, b^{[1]}} Z^{[1]} \xrightarrow[BN]{\beta^{[1]}, \gamma^{[1]}} \tilde{Z}^{[1]} \rightarrow g^{[1]}(\tilde{Z}^{[1]}) = a^{[1]} \xrightarrow{W^{[2]}, b^{[2]}} Z^{[2]} \rightarrow \dots$$

$$\boxed{X^{\{2\}}} \longrightarrow Z^{[1]} \xrightarrow[BN]{\beta^{[1]}, \gamma^{[1]}} \tilde{Z}^{[1]} \rightarrow \dots$$

$$X^{\{3\}} \longrightarrow \dots$$

Parameters: $W^{[l]}, \cancel{b^{[l]}}, \beta^{[l]}, \gamma^{[l]}$.

$(n^{[l]}, 1) \quad (n^{[l]}, 1) \quad (n^{[l]}, 1)$

$Z^{[l]}$

$(n^{[l]}, 1)$

$$\rightarrow Z^{[l]} = W^{[l]} a^{[l-1]} + \boxed{\cancel{b^{[l]}}}$$

$$Z^{[l]} = W^{[l]} a^{[l-1]}$$

$Z^{[l]}_{norm}$

$$\rightarrow \tilde{Z}^{[l]} = \gamma^{[l]} Z^{[l]}_{norm} + \boxed{\beta^{[l]}}$$

Andrew Ng

Implementing gradient descent

for $t = 1 \ldots$ numMiniBatches

Compute forward prop on $X^{\{t\}}$.

In each hidden layer, use BN to replace $\underline{z^{[l]}}$ with $\underline{\tilde{z}^{[l]}}$.

Use backprop to compute $\underline{dW^{[l]}}$, $\cancel{db^{[l]}}$, $\underline{d\beta^{[l]}}$, $\underline{d\gamma^{[l]}}$

Update parameters

$$\left.\begin{aligned} W^{[l]} &:= W^{[l]} - \alpha\, dW^{[l]} \\ \beta^{[l]} &:= \beta^{[l]} - \alpha\, d\beta^{[l]} \\ \gamma^{[l]} &:= \ldots \end{aligned}\right\} \leftarrow$$

Works w/ momentum, RMSprop, Adam.

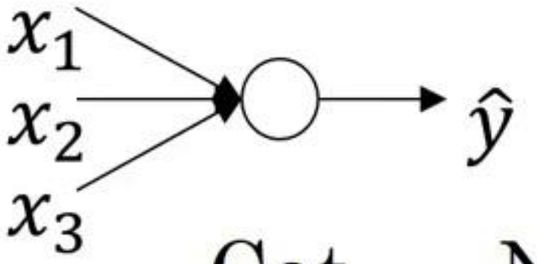
Andrew Ng

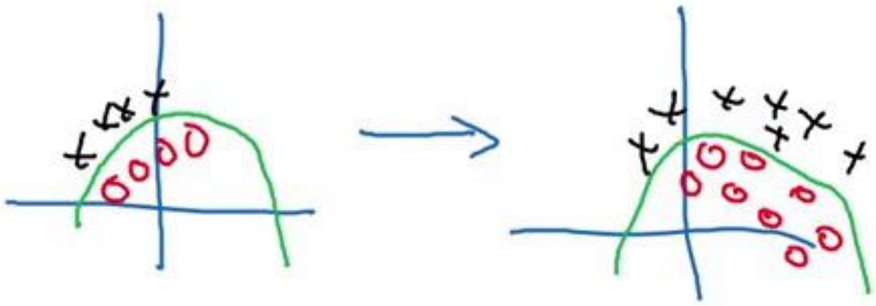
deeplearning.ai

Batch Normalization

Why does Batch Norm work?

Learning on shifting input distribution





Cat $y = 1$

Non-Cat $y = 0$

"Covariate shift"

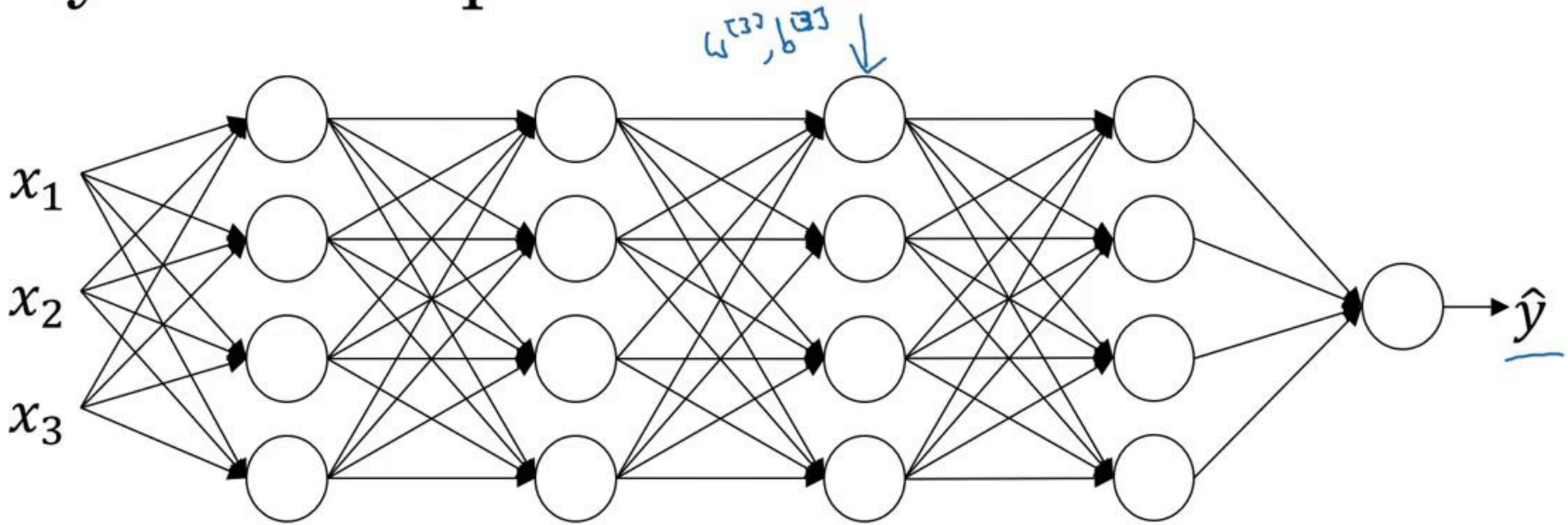
$X \rightarrow y$

$y = 1$

$y = 0$

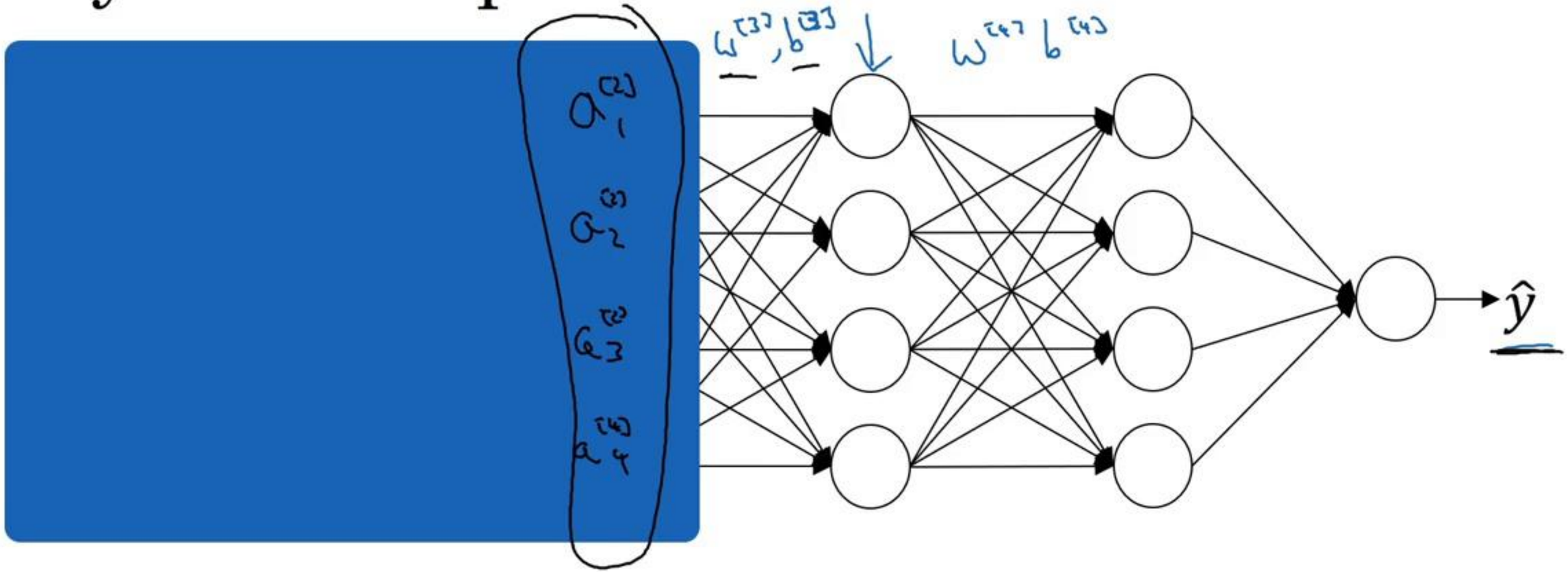
Andrew Ng

Why this is a problem with neural networks?



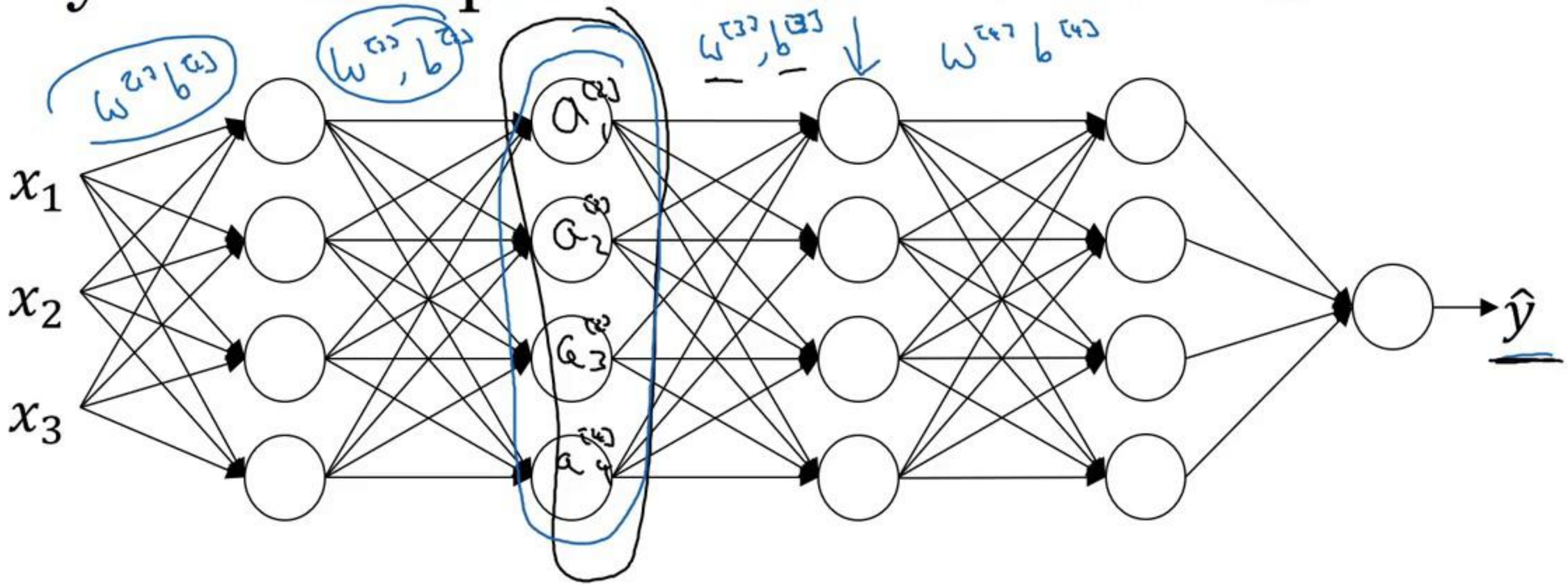
Andrew Ng

Why this is a problem with neural networks?



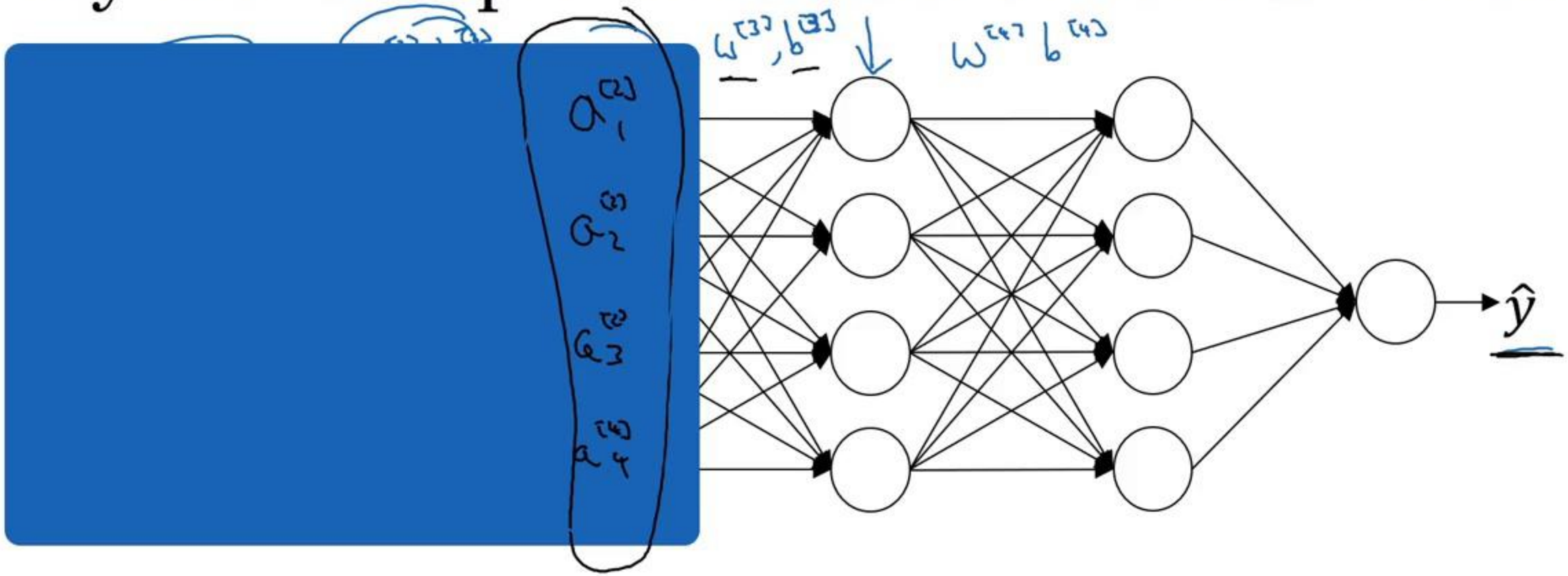
Andrew Ng

Why this is a problem with neural networks?



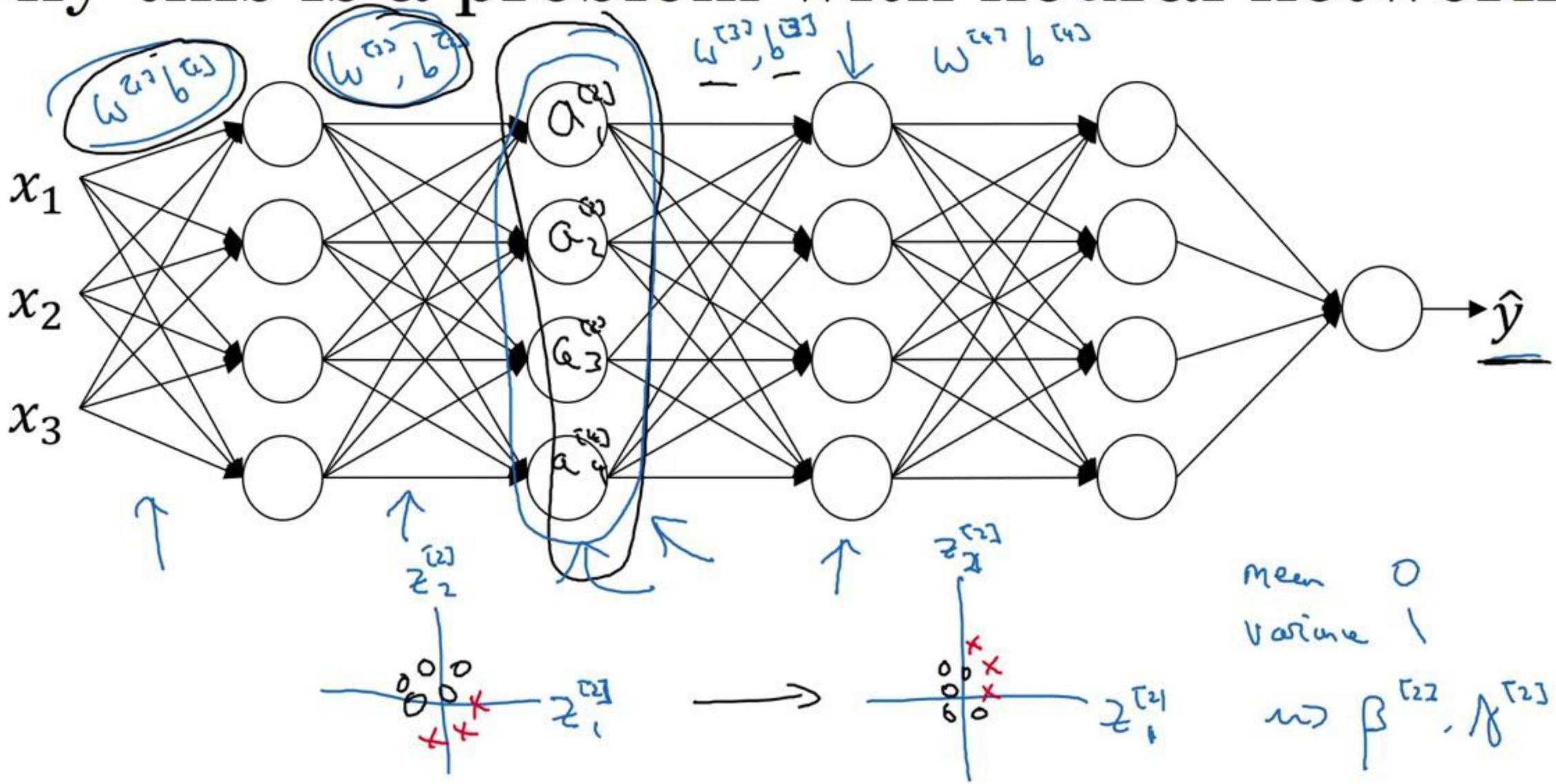
Andrew Ng

Why this is a problem with neural networks?



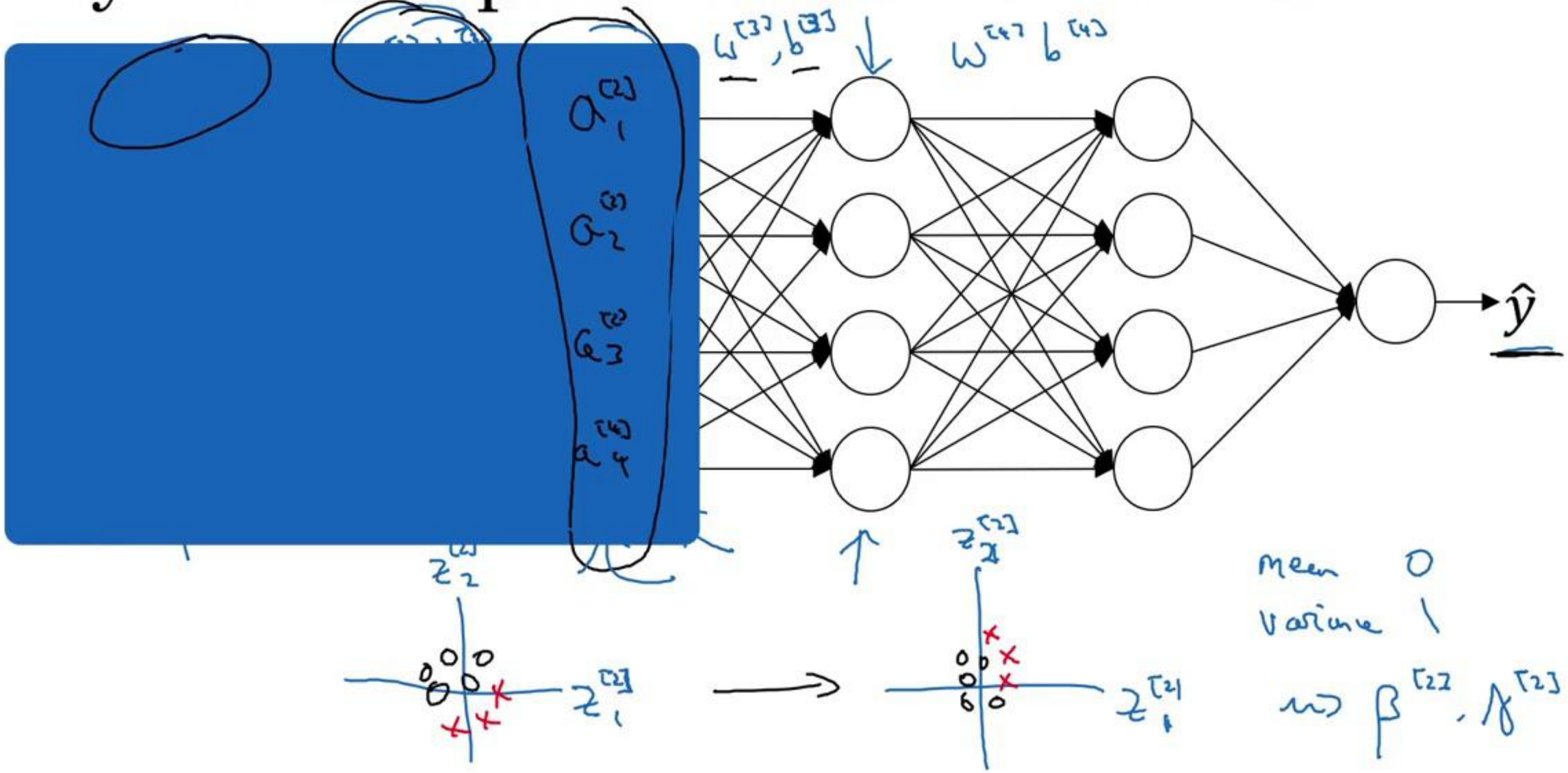
Andrew Ng

Why this is a problem with neural networks?



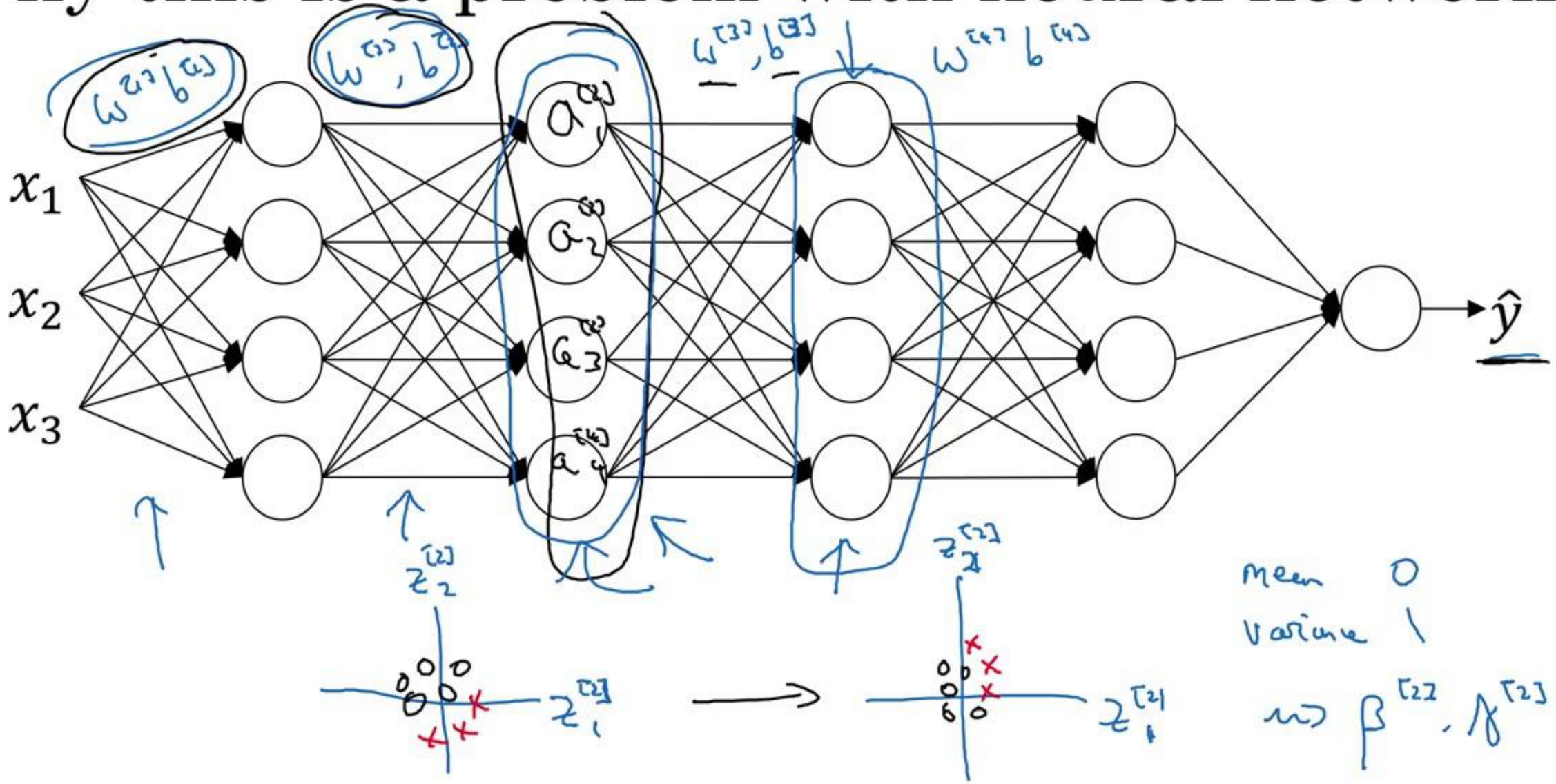
Andrew Ng

Why this is a problem with neural networks?



Andrew Ng

Why this is a problem with neural networks?



Andrew Ng

Batch Norm as regularization

X

$X^{\{t\}}$

- Each mini-batch is scaled by the mean/variance computed on just that mini-batch.

$\tilde{z}^{[l]}$ 64, 128 $Z^{[l]}$

- This adds some noise to the values $z^{[l]}$ within that minibatch. So similar to dropout, it adds some noise to each hidden layer's activations.

μ, σ^2

- This has a slight regularization effect.

mini-batch : 64 $\longrightarrow$ 512

Andrew Ng

deeplearning.ai

Batch Normalization

Batch Norm at test time

Batch Norm at test time

$$\mu = \frac{1}{m}\sum_i z^{(i)}$$

$$\sigma^2 = \frac{1}{m}\sum_i (z^{(i)} - \mu)^2$$

$$z^{(i)}_{\text{norm}} = \frac{z^{(i)} - \mu}{\sqrt{\sigma^2 + \varepsilon}}$$

$$\tilde{z}^{(i)} = \gamma z^{(i)}_{\text{norm}} + \beta$$

μ, σ^2: estimate using exponentially weighted average (across mini-batches).

$X^{\{1\}}, X^{\{2\}}, X^{\{3\}}, \ldots$

$\mu^{[1]}$

$\mu^{\{1\}[1]}$ $\mu^{\{2\}[1]}$ $\mu^{\{3\}[1]}$ $\longrightarrow$ μ

θ_1 θ_2 θ_3 σ^2

$\sigma^{2\{1\}[1]}$ $\sigma^{2\{2\}[1]}$ $\ldots$

$$z_{norm} = \frac{z - \mu}{\sqrt{\sigma^2 + \varepsilon}} \qquad \tilde{z} = \gamma z_{norm} + \beta$$

Andrew Ng

deeplearning.ai

Multi-class classification

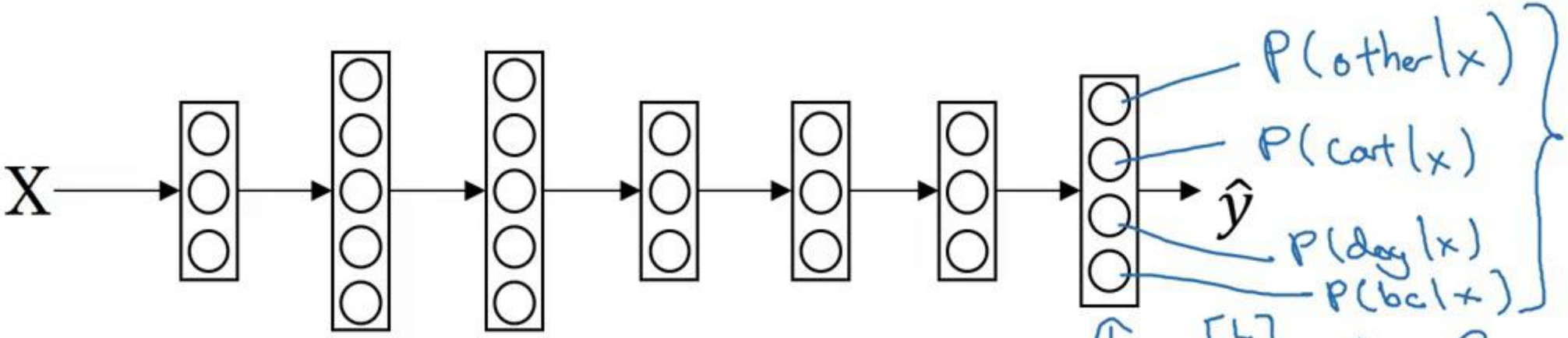
Softmax regression

Recognizing cats, dogs, and baby chicks , other

1 2 3 0

3 1 2 0 3 2 0 1

C = #classes = 4 (0, ..., 3)

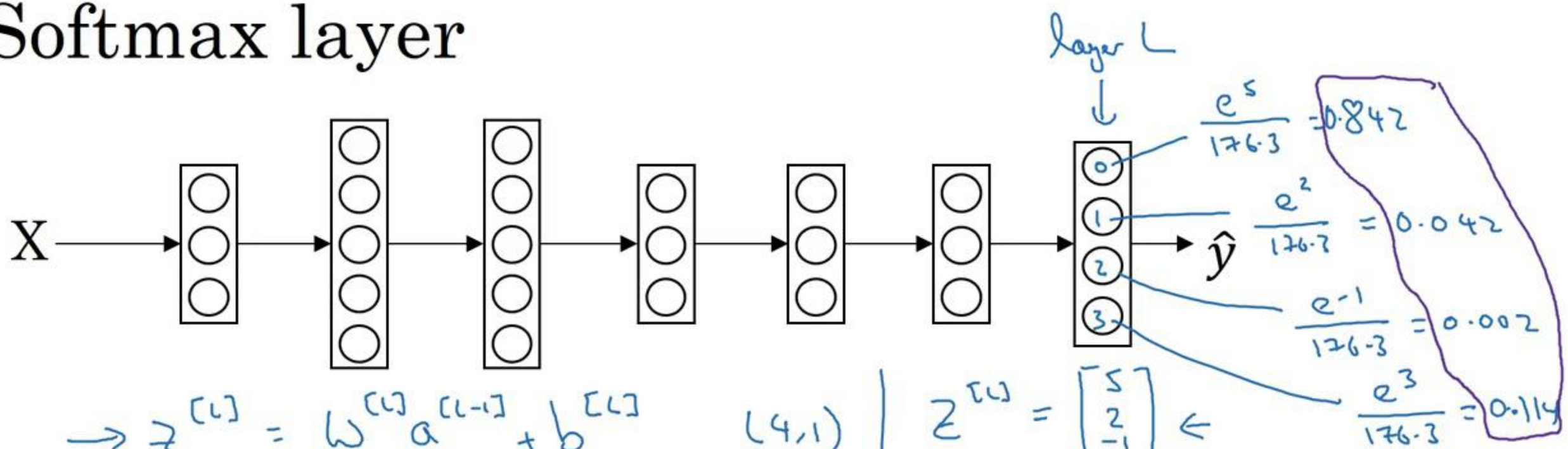


$n^{[L]} = 4 = C$

$\hat{y}$ is (4,1)

Andrew Ng

Softmax layer



$$z^{[L]} = W^{[L]} a^{[L-1]} + b^{[L]} \qquad (4,1)$$

Activation function:

$$t = e^{(z^{[L]})}$$

$$\boxed{a^{[L]}} = \frac{e^{z^{[L]}}}{\sum_{j=1}^{4} t_i}, \quad a_i^{[L]} = \frac{t_i}{\sum_{j=1}^{4} t_i} \qquad (4,1)$$

$\boxed{(4,1)}$ $\hat{y}$

$$\underset{(4,1)}{a^{[L]}} = g^{[L]}(\underset{(4,1)}{z^{[L]}})$$

$$z^{[L]} = \begin{bmatrix} 5 \\ 2 \\ -1 \\ 3 \end{bmatrix}$$

$$t = \begin{bmatrix} e^5 \\ e^2 \\ e^{-1} \\ e^3 \end{bmatrix} = \begin{bmatrix} 148.4 \\ 7.4 \\ 0.4 \\ 20.1 \end{bmatrix}. \quad \sum_{j=1}^{4} t_j = 176.3$$

$$a^{[L]} = \frac{t}{176.3}$$

Andrew Ng

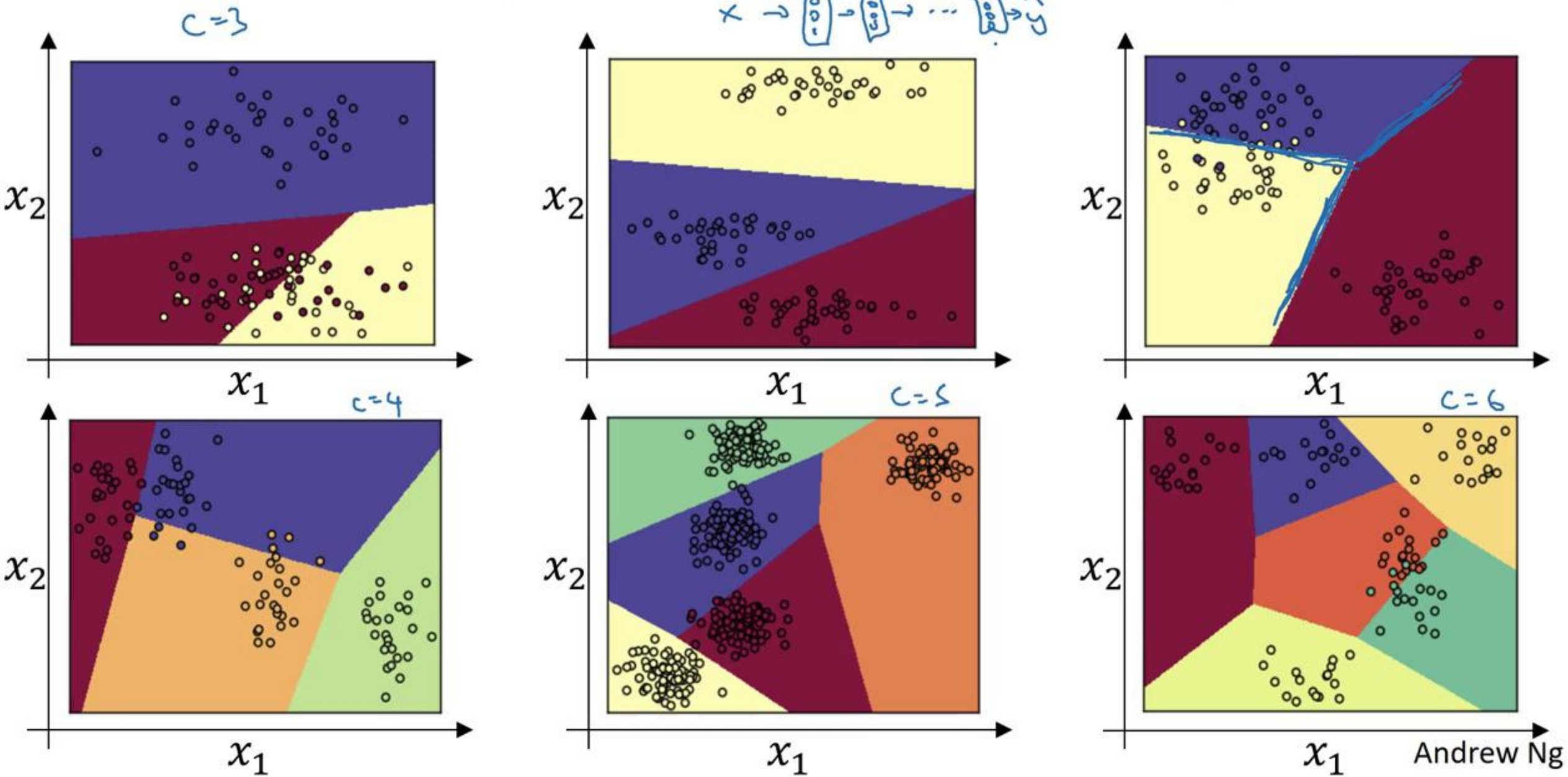
Softmax examples

x_1, x_2 → $\hat{y}$

$z^{[l]} = W^{[l]} x + b^{[l]}$

$a^{[l]} = \hat{y} = g(z^{[l]})$

$x \to \cdots \to \hat{y}$



Andrew Ng

deeplearning.ai

Multi-class classification

Training a softmax classifier

Understanding softmax

(4,1)

$$z^{[L]} = \begin{bmatrix} 5 \\ 2 \\ -1 \\ 3 \end{bmatrix} \qquad \mathrm{t} = \begin{bmatrix} e^5 \\ e^2 \\ e^{-1} \\ e^3 \end{bmatrix}$$

C = 4 $\qquad g^{[L]}(\cdot)$

"soft max"

$$a^{[L]} = g^{[L]}(z^{[L]}) = \begin{bmatrix} e^5/(e^5+e^2+e^{-1}+e^3) \\ e^2/(e^5+e^2+e^{-1}+e^3) \\ e^{-1}/(e^5+e^2+e^{-1}+e^3) \\ e^3/(e^5+e^2+e^{-1}+e^3) \end{bmatrix} = \begin{bmatrix} 0.842 \\ 0.042 \\ 0.002 \\ 0.114 \end{bmatrix}$$

"hard max"

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Softmax regression generalizes logistic regression to C classes.

If $\underline{C=2}$, softmax reduces to logistic regression. $a^{[L]} = \begin{bmatrix} 0.842 \\ \cancel{0.158} \end{bmatrix}$

Andrew Ng

Loss function

$$\overset{(4,1)}{y^{(i)}} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \begin{matrix} \\ \leftarrow \text{cat} \\ \\ \\ \end{matrix} \qquad y_2 = 1$$

$$y_1 = y_3 = y_4 = 0$$

$$a^{[L](i)} = \hat{y}^{(i)} = \overset{(4,1)}{\begin{bmatrix} 0.3 \\ 0.2 \\ 0.1 \\ 0.4 \end{bmatrix}} \qquad C = 4$$

$$\underbrace{\mathcal{L}(\hat{y}, y)}_{\text{small}} = \underbrace{-\sum_{j=1}^{4} y_j \log \hat{y}_j}$$

$$J(w^{[1]}, b^{[1]}, \ldots) = \frac{1}{m} \sum_{i=1}^{m} \mathcal{L}(\hat{y}^{(i)}, y^{(i)})$$

$$-y_2 \log \hat{y}_2 = \underline{-\log \hat{y}_2}.$$

Make $\hat{y}_2$ big.

$$Y = [y^{(1)}\ y^{(2)} \ldots, y^{(m)}] = \begin{bmatrix} 0 & 0 & 1 & \\ 1 & 0 & 0 & \cdots \\ 0 & 1 & 0 & \\ 0 & 0 & 0 & \end{bmatrix}$$

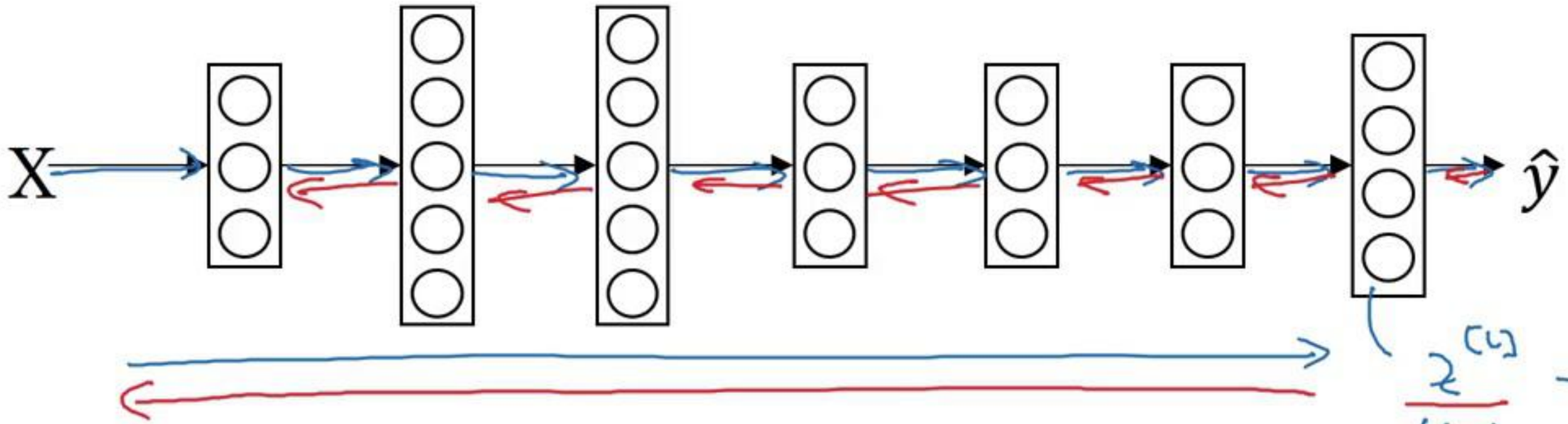
$(4, m)$

$$\hat{Y} = [\hat{y}^{(1)}, \ldots, y^{(m)}] = \begin{bmatrix} 0.3 & \\ 0.2 & \cdots \\ 0.1 & \\ 0.4 & \end{bmatrix}$$

$(4, m)$

Andrew Ng

Gradient descent with softmax



$\underset{(4,1)}{z^{[L]}} \longrightarrow a^{[L]} = \hat{y} \rightarrow \mathcal{L}(\hat{y}, y)$

Backprop:

$$\underset{(4,1)}{dz^{[L]}} = \hat{y} - y$$

$$\frac{\partial J}{\partial z^{[L]}}$$

Andrew Ng

deeplearning.ai

Programming Frameworks

Deep Learning frameworks

Deep learning frameworks

- Caffe/Caffe2
- CNTK
- DL4J
- Keras
- Lasagne
- mxnet
- PaddlePaddle
- TensorFlow
- Theano
- Torch

Choosing deep learning frameworks

- Ease of programming (development and deployment)
- Running speed
- Truly open (open source with good governance)

Andrew Ng

deeplearning.ai

Programming Frameworks

TensorFlow

Motivating problem

$J(w) = \boxed{w^2 - 10w + 25}$

(cost)

$(w-5)^2$

$w = 5$

$J(w, b)$

TensorFlow example

localhost:8891/notebooks/TensorFlow%20example%20.ipynb

jupyter TensorFlow example Last Checkpoint: 5 minutes ago (unsaved changes)

Logout

File Edit View Insert Cell Kernel Help

Trusted Python 3

Code

In [1]:
```
import numpy as np
import tensorflow as tf
```

In [2]:
```
w = tf.Variable(0,dtype=tf.float32)
cost = tf.add(tf.add(w**2,tf.multiply(-10.,w)),25)
train = tf.train.GradientDescentOptimizer(0.01).minimize(cost)

init = tf.global_variables_initializer()
session = tf.Session()
session.run(init)
print(session.run(w))
```

```
0.0
```

In [3]:
```
session.run(train)
print(session.run(w))
```

```
0.1
```

In [4]:
```
for i in range(1000):
    session.run(train)
print(session.run(w))
```

```
4.99999
```

TensorFlow example

localhost:8891/notebooks/TensorFlow%20example%20.ipynb

jupyter TensorFlow example Last Checkpoint: 8 minutes ago (unsaved changes)

Logout

File Edit View Insert Cell Kernel Help Trusted Python 3

Code

In [1]:
```
import numpy as np
import tensorflow as tf
```

In [5]:
```
w = tf.Variable(0,dtype=tf.float32)
x = tf.placeholder(tf.float32, [3,1])
#cost = tf.add(tf.add(w**2,tf.multiply(-10.,w)),25)
cost = w**2 - 10*w + 25
train = tf.train.GradientDescentOptimizer(0.01).minimize(cost)

init = tf.global_variables_initializer()
session = tf.Session()
session.run(init)
print(session.run(w))
```

```
0.0
```

In [6]:
```
session.run(train)
print(session.run(w))
```

```
0.1
```

In [7]:
```
for i in range(1000):
    session.run(train)
print(session.run(w))
```

```
4.99999
```

TensorFlow example

localhost:8891/notebooks/TensorFlow%20example%20.ipynb

jupyter TensorFlow example Last Checkpoint: 10 minutes ago (unsaved changes)

Logout

File Edit View Insert Cell Kernel Help

Trusted Python 3

Code

In [1]:
```
import numpy as np
import tensorflow as tf
```

In [8]:
```
coefficients = np.array([[1.], [-20.], [100.]])

w = tf.Variable(0,dtype=tf.float32)
x = tf.placeholder(tf.float32, [3,1])
#cost = tf.add(tf.add(w**2,tf.multiply(-10.,w)),25)
#cost = w**2 - 10*w + 25
cost = x[0][0]*w**2 + x[1][0]*w + x[2][0]
train = tf.train.GradientDescentOptimizer(0.01).minimize(cost)

init = tf.global_variables_initializer()
session = tf.Session()
session.run(init)
print(session.run(w))
```

```
0.0
```

In [9]:
```
session.run(train, feed_dict={x:coefficients})
print(session.run(w))
```

```
0.1
```

In [10]:
```
for i in range(1000):
    session.run(train, feed_dict={x:coefficients})
print(session.run(w))
```

TensorFlow example

localhost:8891/notebooks/TensorFlow%20example%20.ipynb

jupyter TensorFlow example Last Checkpoint: 12 minutes ago (autosaved)

Logout

File Edit View Insert Cell Kernel Help

Trusted Python 3

Code

In [1]:
```
import numpy as np
import tensorflow as tf
```

In [11]:
```
coefficients = np.array([[1.], [-20.], [100.]])

w = tf.Variable(0,dtype=tf.float32)
x = tf.placeholder(tf.float32, [3,1])
#cost = tf.add(tf.add(w**2,tf.multiply(-10.,w)),25)
#cost = w**2 - 10*w + 25
cost = x[0][0]*w**2 + x[1][0]*w + x[2][0]
train = tf.train.GradientDescentOptimizer(0.01).minimize(cost)

init = tf.global_variables_initializer()
session = tf.Session()
session.run(init)
print(session.run(w))
```

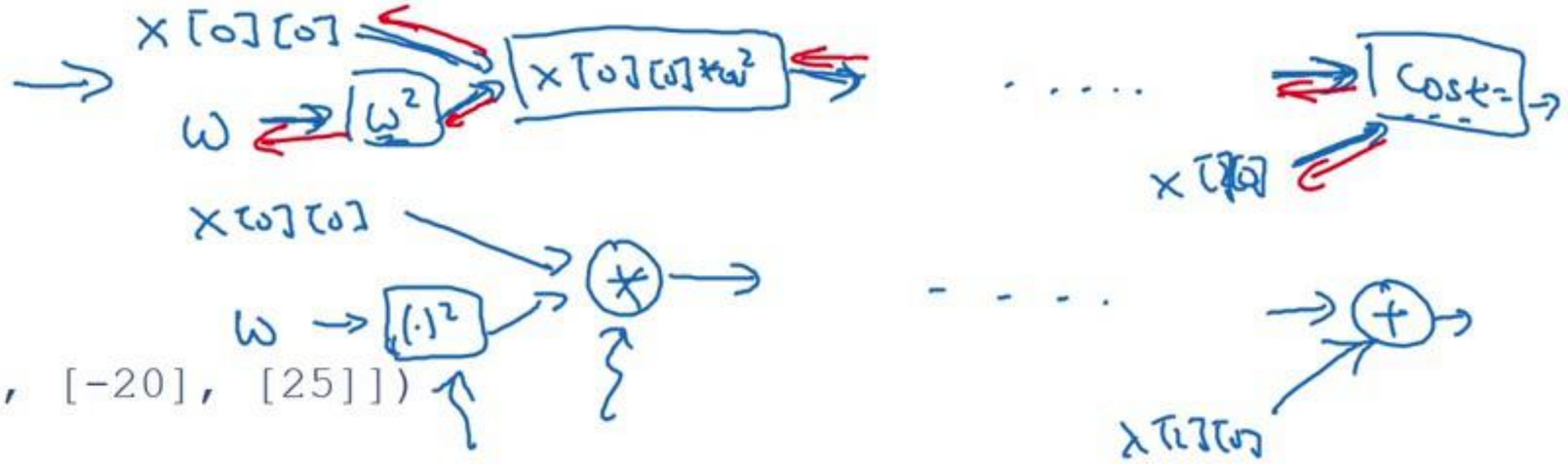
```
0.0
```

In [12]:
```
session.run(train, feed_dict={x:coefficients})
print(session.run(w))
```

```
0.2
```

In [13]:
```
for i in range(1000):
```

Code example




```
import numpy as np
import tensorflow as tf

coefficients = np.array([[1], [-20], [25]])

w = tf.Variable([0],dtype=tf.float32)
x = tf.placeholder(tf.float32, [3,1])
cost = x[0][0]*w**2 + x[1][0]*w + x[2][0]     # (w-5)**2
train = tf.train.GradientDescentOptimizer(0.01).minimize(cost)
init = tf.global_variables_initializer()
session = tf.Session()
session.run(init)
print(session.run(w))

for i in range(1000):
    session.run(train, feed_dict={x:coefficients})
print(session.run(w))
```

```
with tf.Session() as session:
    session.run(init)
    print(session.run(w))
```

Andrew Ng